MECH 191 — Metallurgy: Key Terms & Vocabulary Guide

Terms to notate during lecture and research further — organized by unit and week

MECH 191 — Metallurgy

Leeward Community College · Math and Science Division · Mechatronics Program

Compiled from course lecture slides and reference pages

How to use this guide

This list follows the course roadmap — five units, each spanning several weeks. For every week, the terms below are the vocabulary worth writing down during lecture (on your note-taking handout) and looking up further if the in-class explanation leaves you wanting more depth. A short working definition is included for context, but the real value is in researching each term further: find a diagram, a real-world failure example, or the relevant code/standard that governs it.

A few terms repeat across weeks (e.g., grain boundaries, sensitization) — that repetition is intentional in the course design, since Unit 5 explicitly reconnects earlier concepts to failure and corrosion.

A "Units of Measurement & Dimensional Analysis" section comes next — it starts from the seven SI base units, shows how dimensional analysis builds every derived unit (force, stress, energy, power) used in this course from just length, mass, time, and temperature, walks through a few worked examples checking the units a formula produces, and closes with the U.S. customary units (ksi, ft-lbf, °F, mils) still used alongside SI on the shop floor.

A "Quick Math Review" follows that — order of operations, fractions and ratios, percentages, exponents, square roots, scientific notation and log scales, and rearranging an equation to solve for an unknown — each rule illustrated with a worked example pulled directly from a formula elsewhere in this guide (the carbon equivalent formula, Hall-Petch, the Taylor tool life equation, the Brinell hardness formula, and others).

A short "Geometry & Right-Triangle Trigonometry" refresher comes next — angles, circle area and circumference, the Pythagorean theorem, and the sine/cosine/tangent ratios (SOH-CAH-TOA) — grounded in two real course calculations: the 0.707 fillet weld throat factor and the depth/surface-distance trig used to locate a flaw with an angle-beam ultrasonic probe.

A final section, "Engineering Equations & Key Concepts," collects the formulas used across the course, grouped by topic rather than by week since they resurface throughout the semester: stress/strain and true stress-strain relationships, stress concentration and notch effects (Kt, Kf, notch sensitivity), hardness number calculations, fatigue and impact energy, manufacturing equations for cutting mechanics and tool life, forming and sheet-metal relationships (extrusion ratio, forming limit diagrams), weldability indices (CE, Pcm), corrosion (PREN), and NDT physics.

Unit 1 — Materials Fundamentals

Week 1 — Atomic Structure & Bonding

Metallurgy	The science of metals — structure, properties, and behavior.
Alloy	An intentional mixture of two or more elements engineered for desired properties.
Proton / Neutron / Electron	The three subatomic particles; protons define the element, neutrons define isotopes, electrons govern bonding.
Valence electrons	Outermost electrons — the only ones involved in bonding.
Periodic table groups & periods	Columns (groups) share bonding behavior; rows (periods) add electron shells.
Metallic bond	Electrons form a shared 'electron sea' — explains conductivity and ductility.
Ionic bond	Electron transfer between metal and non-metal; high melting point, brittle.
Covalent bond	Shared electrons between non-metals; strong, directional bonds.
Octet rule	Atoms bond to reach 8 stable valence electrons.
Substitutional solid solution	Alloying atoms of similar size replace host atoms in the lattice.
Interstitial solid solution	Small alloying atoms fit into gaps between host atoms (e.g., carbon in iron).
Unit cell	Smallest repeating building block of a crystal structure.
Simple cubic (SC)	A crystal structure with one atom at each corner of a cube, touching along the cube edges; the loosest-packed of the four common structures (~52% packing factor) and rare among metals.
Body-centered cubic (BCC)	A crystal structure with atoms at the eight corners of a cube plus one atom in the center; moderately packed (~68%), found in ferrite (α-iron), chromium, and tungsten — generally more brittle at low temperature than FCC metals.
Face-centered cubic (FCC)	A crystal structure with atoms at the eight corners of a cube plus one atom centered on each of the six faces; densely packed (~74%), found in austenite (γ-iron), aluminum, copper, and nickel — generally the most ductile common structure.
Hexagonal close-packed (HCP)	A crystal structure built from stacked hexagonal layers of atoms; densely packed (~74%) but with fewer active slip systems than FCC, found in zinc, magnesium, and titanium — generally the least ductile of the four common structures.
Grain / grain boundary	A single crystal domain; the boundary between differently oriented grains.
Polycrystalline	Metal composed of many grains with different orientations.
Isotope	An atom of a given element that has the same number of protons but a different number of neutrons, giving it a different atomic mass while keeping the same chemical behavior.

Electron shell	A discrete energy level surrounding the nucleus where electrons orbit; each successive period (row) of the periodic table adds one additional shell.
Crystal lattice	The repeating, three-dimensional geometric arrangement of atoms (or lattice points) that defines the structure of a crystalline solid; alloying atoms and defects occupy specific sites within this arrangement.
Atomic packing factor	The fraction of a unit cell's volume occupied by atoms modeled as hard spheres (APF = volume of atoms in the cell / total cell volume); it measures how densely packed a crystal structure is, e.g., ~0.68 for BCC versus ~0.74 for FCC and HCP.

Week 2 — Mechanical Properties & Behavior

Stress (σ) / Strain (ε)	Internal force per unit area; deformation relative to original size.
Tensile stress	A stress that pulls a material apart along its axis, stretching it in the direction of the applied load.
Compressive stress	A stress that pushes a material together along its axis, squeezing and shortening it in the direction of the applied load.
Shear stress	A stress caused by forces acting parallel to a surface rather than perpendicular to it, causing adjacent layers of material to slide relative to one another.
Elastic deformation vs. plastic deformation	Recoverable vs. permanent deformation.
Yield point / yield strength	Stress at onset of permanent plastic deformation.
Ultimate tensile strength (UTS)	Maximum stress on the stress-strain curve, where necking begins.
Strength	A material's ability to resist an applied load without yielding or fracturing; most commonly reported as yield strength (the stress at which permanent deformation first begins) or ultimate tensile strength (the maximum stress the material reaches before necking).
Hardness	A material's resistance to localized permanent (plastic) deformation at its surface, measured by pressing a standardized indenter into the material under a known load and reading the size or depth of the resulting impression (Rockwell, Brinell, Vickers).
Ductility	A material's ability to undergo large plastic deformation — especially stretching — before it fractures, reported from a tensile test as % elongation or % reduction in area; high-ductility metals can be drawn into wire or deep-drawn into sheet parts.
Toughness	A material's ability to absorb energy and deform plastically before fracturing, combining strength and ductility; measured as the area under a material's stress-strain curve, or directly with an impact test (Charpy/Izod).
Brittleness	A material's tendency to fracture with little or no plastic deformation once its strength is exceeded — the opposite of ductility and toughness; brittle fractures happen suddenly and leave a flat, crystalline surface rather than necking or dimpling.
Point defect: vacancy, interstitial, substitutional	Missing atom; extra squeezed-in atom; wrong atom in a lattice site.
Dislocation	A line defect in the crystal lattice that lets a metal deform plastically by shifting atoms into new bonds without breaking every bond in the crystal at once; described by its Burgers vector.
Edge dislocation	A line defect formed where an extra partial plane of atoms terminates inside the crystal, creating a step in the lattice; it moves by atoms shuffling one position at a time along the slip plane, roughly in the direction of the applied stress.

Screw dislocation	A line defect in which the lattice planes wind around the dislocation line like a spiral staircase; it moves by a shearing motion parallel to the dislocation line itself, distinct from how an edge dislocation moves.
Mixed dislocation	A dislocation that is part edge and part screw character along different segments of its length — the form most dislocations actually take in real metals.
Grain boundary (planar defect)	Region where crystal orientation changes; high vs. low misorientation.
Twin boundary (annealing vs. deformation twins)	Mirror-image boundary; low energy, very stable.
Stacking fault	Disruption in the normal atomic stacking sequence; relates to stacking fault energy (SFE).
Void	An empty cavity or gap within an otherwise solid material, arising from trapped gas, shrinkage during solidification, or incomplete fusion.
Inclusion	A foreign, non-metallic particle (such as oxide, sulfide, or slag) trapped inside a metal during solidification or processing, acting as a stress concentrator and potential crack-initiation site.
Delamination	A planar separation between layers within a material, running roughly parallel to the surface.
Segregation	A non-uniform, localized concentration of an alloying element or impurity that differs from the material's average composition, typically arising during solidification.
Crack	A narrow fracture surface within a material where the base metal has separated, initiated either during processing (e.g. from residual stress or hydrogen) or in service (e.g. from fatigue or corrosion).
Hall-Petch relationship	$\sigma_\gamma = \sigma_0 + k/\sqrt{d}$ — smaller grain size increases yield strength.
Recovery	The first stage of annealing after cold work, in which internal stresses relax and dislocations rearrange into lower-energy configurations, restoring some ductility with only a small drop in strength and no change in grain structure.
Recrystallization	The second stage of annealing, in which new, strain-free grains nucleate and grow to replace the deformed grain structure, substantially restoring ductility while dropping strength back toward the annealed condition.
Grain growth	The third stage of annealing, in which recrystallized grains continue to grow — larger grains consuming smaller ones — if held at temperature too long, which can reduce strength since grain-boundary strengthening decreases as grains coarsen.
Diffusion (substitutional vs. interstitial)	Atomic movement through a solid; rate increases exponentially with temperature.
Work hardening	Strengthening from dislocation accumulation during cold work.
Fatigue / endurance limit	Failure below yield strength from repeated cyclic loading.

Necking	The localized reduction in cross-sectional area that begins once a ductile material reaches its ultimate tensile strength, concentrating further plastic deformation in that region until fracture.
Burgers vector	A vector that defines the magnitude and direction of lattice distortion caused by a dislocation, found by tracing a closed loop around the dislocation line and measuring the closure failure compared to a perfect lattice.
Stacking fault energy (SFE)	The energy per unit area associated with a stacking fault; metals with low SFE form wide faults and deform more readily by twinning, while high-SFE metals recover more easily through dislocation cross-slip.
Fracture toughness	A material property, commonly denoted K_IC, that quantifies a material's resistance to crack propagation from an existing flaw; a higher fracture toughness means the material can tolerate a larger crack or higher stress before it fractures suddenly.
Flow stress	The true stress needed to keep a metal deforming plastically at a given strain. Per the Hollomon equation $\sigma = K\varepsilon^n$, flow stress rises with strain but more gradually as the strain-hardening exponent n gets smaller.
Elastic modulus (Young's modulus)	The slope of the linear (elastic) portion of the stress-strain curve, representing a material's stiffness — its resistance to elastic (recoverable) deformation under load. Reported in GPa or psi; steel (~200 GPa) is roughly three times stiffer than aluminum (~70 GPa).
Proportional limit	The highest stress at which stress and strain remain linearly related (Hooke's Law still applies); beyond this point the stress-strain curve begins to curve and permanent deformation starts to accumulate.

Unit 2 — Steel & Alloys

Week 4 — Carbon & Alloy Steels

Ferrite / austenite (BCC/FCC iron)	Low-temp BCC iron phase vs. high-temp FCC iron phase — the basis of steel allotropy.
Carbon equivalent (CE)	CE = %C + %Mn/6 + (%Cr+%Mo+%V)/5 + (%Ni+%Cu)/15 — determines weldability/preheat needs.
Low carbon steel	Steel with roughly 0.05-0.30% carbon; soft, ductile, and easily welded and formed, but not very hardenable by heat treatment — typical structural and sheet-metal grades.
Medium carbon steel	Steel with roughly 0.30-0.60% carbon; a balance of strength and ductility that responds well to heat treatment, commonly used for shafts, gears, and machine parts.
High carbon steel	Steel with roughly 0.60-1.00%+ carbon; high strength and hardness after heat treatment but reduced ductility and weldability, used for tools, springs, and cutting edges.
AISI/SAE four-digit designation	First two digits = steel family/alloy; last two = carbon content in hundredths of a percent.
Manganese (Mn)	An alloying element that improves hardenability and strength and ties up sulfur (as MnS) to prevent hot shortness during hot working.
Chromium (Cr)	An alloying element that improves hardenability and, at higher levels (≥10.5%), provides corrosion resistance by forming a passive chromium-oxide layer — the basis of stainless steel.
Nickel (Ni)	An alloying element that improves toughness and low-temperature impact resistance, and stabilizes the austenite (FCC) phase in stainless steels.
Molybdenum (Mo)	An alloying element that improves hardenability and high-temperature strength and resists temper embrittlement during tempering.
Vanadium (V)	An alloying element that refines grain size and forms hard carbides, improving strength, wear resistance, and temper resistance.
Silicon (Si)	An alloying element that acts primarily as a deoxidizer during steelmaking and increases strength in spring steels and electrical (silicon) steels.
Hardenability	How deeply a steel can be hardened through its cross-section (different from hardness).
CCT diagram (Continuous Cooling Transformation)	Plots temperature vs. time to show what microstructure actually forms during real cooling.
Ms / Mf (martensite start/finish)	Temperatures at which martensite begins and finishes forming.
Pearlite	A layered (lamellar) microstructure of alternating ferrite and cementite plates that forms from austenite under moderate cooling rates; moderate strength and ductility.

Bainite	A fine, feathery to needle-like microstructure of ferrite and cementite that forms at cooling rates between those producing pearlite and martensite; offers a good combination of strength and toughness.
Martensite	A hard, brittle, supersaturated body-centered tetragonal microstructure that forms when austenite is cooled too fast for diffusion to occur, trapping carbon in the iron lattice; requires tempering to restore usable toughness.
Annealing	Heat to austenite, furnace cool — softens and restores ductility.
Normalizing	Heat to austenite, air cool — refines grain size.
Quenching	Rapid cool from austenite — maximizes hardness (forms martensite).
Tempering	Reheating quenched martensite — restores toughness, reduces hardness/brittleness.
Case hardening	A family of heat-treatment processes that hardens only a component's surface layer, leaving a tough, ductile core underneath.
Carburizing	A case-hardening process that diffuses carbon into a low-carbon steel's surface at high temperature, followed by quenching, creating a hard, wear-resistant surface layer over a tough, ductile core.
Nitriding	A case-hardening process that diffuses nitrogen into a steel's surface at lower temperature — forming hard nitride compounds — without requiring a subsequent quench, giving a hard case with minimal distortion.
Post-weld heat treatment (PWHT)	Heat treatment after welding to relieve stress and temper the HAZ.
Allotropy	Allotropy is the ability of a solid element to exist in more than one crystal structure depending on temperature or pressure. Iron is allotropic, shifting between BCC (ferrite) at low temperature and FCC (austenite) at high temperature.
Weldability	Weldability is the ease with which a metal can be welded without cracking, porosity, or the need for extensive preheat and post-weld heat treatment. It generally decreases as carbon content and alloying (higher carbon equivalent) increase.
Microstructure	Microstructure is the arrangement, size, and distribution of grains and phases within a metal, visible under magnification. It is controlled by composition and thermal history, and it governs mechanical properties such as hardness, strength, and toughness.
Cooling rate	The rate at which a steel drops in temperature after heating, typically expressed in °C/s. It is the key variable read off a CCT diagram, since a faster cooling rate suppresses pearlite/bainite formation and favors martensite.
HSLA steel	High-Strength, Low-Alloy steel — carbon steel (typically under about 0.18% C) strengthened primarily through small additions of alloying elements (like Nb, V, Ti) and grain refinement rather than through high carbon content, giving good strength while retaining weldability.

Week 5 — Stainless Steels & Cast Irons

Passive chromium oxide layer (Cr_2O_3)	Self-repairing surface film that makes stainless steel corrosion-resistant; requires ≥10.5% Cr.
Austenitic stainless (300 series)	FCC, non-magnetic, not hardenable by heat treatment — e.g. 304, 316.
Ferritic stainless (400 series)	BCC, magnetic, not hardenable — e.g. 430.
Martensitic stainless (400 series)	BCC/BCT, magnetic, hardenable by quench & temper — e.g. 410, 420, 440C.
Duplex stainless	Mixed austenite + ferrite microstructure — higher strength, better SCC resistance.
Sensitization	Chromium carbide precipitation at grain boundaries during welding — depletes nearby Cr, causing intergranular corrosion susceptibility.
L-grade (304L, 316L)	Low-carbon grades used to prevent sensitization in welded fabrication.
Pitting / crevice corrosion / stress corrosion cracking (SCC)	Localized corrosion mechanisms especially relevant to stainless in chloride environments.
Cast iron (definition)	Iron-carbon alloy >2.14% carbon, typically 2.5-4%, plus 1-3% silicon.
Gray cast iron	Carbon as graphite flakes — good damping/machinability, low tensile strength.
White cast iron	Carbon as iron carbide/cementite — extremely hard, brittle, wear-resistant.
Ductile (nodular) iron	Mg/Ce addition makes graphite form as spheroids — much tougher than gray iron.
Malleable cast iron	White iron annealed to convert cementite to temper carbon rosettes.
Cementite (Fe_3C)	Iron carbide — extremely hard, brittle phase formed above carbon solubility limit.
BCT (Body-Centered Tetragonal)	BCT is a crystal structure like BCC but stretched along one axis. Martensite adopts this structure because carbon atoms trapped in the lattice during rapid quenching distort it, producing high hardness and internal strain.
Intergranular corrosion	Intergranular corrosion is localized attack that proceeds along grain boundaries rather than across the grain faces. It commonly results from chromium-depleted zones at the boundaries, which lose the protective passive oxide layer, such as after sensitization.
Graphite	Graphite is the soft, stable, carbon-rich phase that precipitates out of cast iron instead of remaining combined as cementite. Its shape — flakes, spheroids, or rosettes — determines whether the iron is classified as gray, ductile, or malleable.
Temper carbon (rosettes)	Temper carbon rosettes are irregular, cluster-shaped graphite particles that form when the cementite in white cast iron decomposes during prolonged annealing. Their presence gives malleable iron much better ductility and

	toughness than white iron.
Machinability	Machinability is a material's relative ease of being cut, drilled, or machined to a good surface finish, with reasonable cutting forces, low tool wear, and manageable chip formation.

Week 6 — Nonferrous Metals & Phase Diagrams

Aluminum alloy series (1xxx–7xxx)	Designation system by principal alloying element; heat-treatable vs. work-hardened series.
T6 temper	Solution treat + age — most common heat-treated condition for 6061/7075 aluminum.
Brass (Cu-Zn) vs. Bronze (Cu-Sn / Cu-Al)	The two major copper alloy families — know composition and typical uses.
Alpha, beta, alpha-beta titanium alloys	HCP, BCC, and mixed titanium microstructures; Ti-6Al-4V is the workhorse alpha-beta alloy.
Iron-carbon phase diagram	Maps stable phases across carbon content and temperature — ferrite, austenite, cementite, pearlite.
Eutectoid point	0.77% C at 727°C — austenite transforms simultaneously to ferrite + cementite.
Hypoeutectoid steel	Steel with less than 0.77% carbon, which on slow cooling forms a mixture of ferrite and pearlite.
Eutectoid steel	Steel at exactly 0.77% carbon, which on slow cooling transforms entirely to pearlite with no separate ferrite or cementite regions.
Hypereutectoid steel	Steel with more than 0.77% carbon, which on slow cooling forms a mixture of cementite and pearlite.
Galvanic risk / anodic-cathodic relationship (Al vs. steel/Cu)	Aluminum is anodic to steel and copper — avoid direct contact in wet environments.
Solution treatment and aging	Solution treatment and aging is a two-step heat treatment for precipitation-hardenable alloys. The alloy is first heated to dissolve alloying elements into a supersaturated solid solution and quenched (solution treat), then held at a lower temperature so fine strengthening precipitates form (aging).
CP titanium (Commercially Pure)	Unalloyed titanium (ASTM Grades 1-4) containing only trace oxygen, iron, and other interstitial impurities; it is essentially all-alpha phase, offering excellent corrosion resistance and formability but lower strength than the alloyed titanium grades.
Near-alpha titanium alloys	Titanium alloys that are predominantly alpha-phase but contain a small addition (a few percent) of beta-stabilizing elements, giving better high-temperature strength and creep resistance than CP titanium while retaining good weldability and toughness.
Weight percent (wt%) vs. atomic percent (at%)	Two ways of expressing alloy composition: weight percent (wt%) gives each element's share of the total mass, while atomic percent (at%) gives its share of the total number of atoms. Phase diagrams and standard alloy specs are normally read in wt%, so a composition must be converted before comparing it to a mole-based (at%) value.

Unit 3 — Testing, Inspection & Quality

Week 7 — Mechanical Testing Methods

0.2% offset method	Technique for finding yield strength when no clear yield point exists (e.g., aluminum).
Reduction in area / elongation at fracture	Ductility measures from a tensile test.
Rockwell hardness test	Presses a diamond cone or hardened ball indenter into the surface under a minor then major load and reads hardness directly off the depth of penetration; fast and simple, the standard shop-floor check for finished, heat-treated steel parts.
Brinell hardness test	Presses a large hardened steel or carbide ball into the surface under a heavy load and measures the diameter of the resulting impression; well suited to coarse-grained or heterogeneous materials like castings, where an average over a larger area is wanted.
Vickers hardness test	Presses a small pyramidal diamond indenter into the surface and measures the diagonal of the resulting impression; one continuous scale that works across the full hardness range, including thin sections, small parts, and coatings, where Rockwell or Brinell would be too coarse.
Charpy impact test	An impact toughness test in which a notched specimen is supported horizontally at both ends and struck from behind the notch by a swinging pendulum, measuring the energy absorbed in fracture.
Izod impact test	An impact toughness test in which a notched specimen is clamped vertically as a cantilever and struck from the front of the notch by a swinging pendulum, measuring the energy absorbed in fracture.
Ductile-brittle transition temperature (DBTT)	Temperature below which a material's fracture behavior shifts from tough to brittle.
Upper shelf / lower shelf energy	High-energy ductile fracture region vs. low-energy brittle fracture region on an impact-energy-vs-temperature curve.
S-N curve (Wöhler curve)	Plots stress amplitude vs. cycles to failure — used to find the fatigue limit.
Fatigue limit / fatigue ratio	Stress below which steel can survive indefinite cycles; ratio of fatigue limit to UTS.
Mill Test Report (MTR)	Traceability document linking a heat/lot number to material grade, composition, and mechanical test results.
Heat number / lot number	Unique identifier tying a physical piece of material to its test certificate.
Tensile test	A mechanical test in which a specimen is pulled along its axis at a controlled rate until it necks and fractures, while load and elongation are continuously recorded to determine yield strength, ultimate tensile strength, elastic modulus, and ductility.
Indenter (ball indenter)	A hardened tip — a steel or tungsten-carbide ball, diamond cone, or diamond pyramid, depending on the test — that is pressed into a material's

	surface under a known load to leave an impression whose size or depth is converted into a hardness number.
Stress amplitude	Half the range between the maximum and minimum stress applied during one cycle of fatigue loading; it is the quantity plotted on the vertical axis of an S-N (Wöhler) curve.

Week 8 — Non-Destructive Testing (NDT) Methods

Visual testing (VT)	The foundational inspection method — naked eye, lighting, magnification, borescopes.
Liquid/dye penetrant testing (PT)	Capillary action draws penetrant into surface-breaking defects; color contrast vs. fluorescent.
Magnetic particle testing (MT)	Detects surface/near-surface defects in ferromagnetic materials via flux leakage; yoke vs. prod methods.
Ultrasonic testing (UT)	Sound waves detect internal defects; A-scan, phased array (PAUT), straight vs. angle beam.
Radiographic testing (RT)	X-ray/gamma imaging of volumetric defects; film vs. digital radiography.
Eddy current testing (ET)	Induces circulating currents to detect surface cracks/corrosion; governed by skin depth.
Couplant	Gel/oil/water medium between UT transducer and part surface (air poorly transmits ultrasound).
Dwell time	How long penetrant is left on the surface before removal in PT.
Two-direction rule (MT)	Inspections must be performed in at least two perpendicular directions.
ASNT SNT-TC-1A Level I/II/III	NDT technician qualification levels — perform, interpret/accept-reject, and write procedures/certify.
Borescope	A long rigid or flexible optical instrument with a lens, light source, and often a camera at its tip, inserted through small openings to visually inspect internal surfaces and cavities that are otherwise inaccessible to the naked eye.
Capillary action	The tendency of a liquid to be drawn into narrow spaces — such as a surface crack, pore, or gap — without help from external forces, driven by surface tension and the liquid's adhesion to the solid surface it wets.
Ferromagnetic	Describing a material, such as iron, nickel, cobalt, and most carbon and low-alloy steels, that can be strongly magnetized and sustain an induced magnetic field, which is the property that makes it inspectable by magnetic particle testing.
Flux leakage	The escape of magnetic field lines out of a ferromagnetic part at a surface or near-surface discontinuity; the leaking field attracts and holds magnetic particles at that location, revealing the flaw during magnetic particle testing.
Yoke and prod methods (MT magnetization)	Two ways of inducing a magnetic field for magnetic particle testing: a yoke is an electromagnet straddling the part that creates a field between its two poles, while prods are handheld electrodes that pass current directly through the part to set up a circular magnetic field around the current path.
A-scan	The basic ultrasonic testing signal display, plotting echo amplitude (signal strength) on the vertical axis against time, or equivalent depth, on the horizontal axis for a single beam position.
Phased array (PAUT)	An advanced ultrasonic testing technique using a probe made of many small elements that are pulsed individually with precise time delays, allowing the

	sound beam to be steered, focused, and swept electronically without physically moving the probe.
Volumetric defect	A flaw located within the interior thickness of a material or weld, such as porosity, inclusions, or voids, as opposed to a defect that breaks the surface; radiographic and ultrasonic testing are the NDT methods best suited to finding these.
Transducer	The probe used in ultrasonic testing that contains a piezoelectric element, converting electrical pulses into high-frequency sound waves sent into the part and converting the returning echoes back into electrical signals for display.
Electrical conductivity (σ)	A material's ability to carry electric current, the inverse of resistivity. In eddy current testing, higher conductivity induces stronger eddy currents near the surface, giving a smaller skin depth ($\delta = 503/\sqrt{(f \cdot \mu r \cdot \sigma)}$).
Relative permeability (μr)	The ratio of a material's magnetic permeability to that of free space, describing how strongly it concentrates a magnetic field. Ferromagnetic parts have high μr, which sharply reduces eddy current penetration (skin) depth compared to nonmagnetic materials.
Shear-wave velocity	The speed at which a transverse (shear) sound wave propagates through a given material, a property that depends on the material's density and shear modulus; it is used to convert an ultrasonic pulse's round-trip travel time into a physical distance.
Flaw detector	A portable ultrasonic testing instrument that pulses a UT probe, receives the returning echoes, and displays them as amplitude-versus-time signals so an inspector can read off travel time and signal strength.
Probe index point	The marked reference point on an angle-beam UT probe (or its wedge) from which the sound beam is considered to exit; surface distances to a reflector are measured from this point.
Reflector (UT)	Any internal discontinuity or interface — such as a crack, void, inclusion, or the part's back wall — that reflects part of an ultrasonic beam back toward the probe, producing the echo signal read on the flaw detector.
Refracted angle	The angle, measured from the surface normal, at which an ultrasonic beam bends and travels into the part after crossing the wedge-to-part interface; angle-beam probes are built to produce a fixed refracted angle, commonly 45°, 60°, or 70°.

Week 9 — Quality Standards & Inspection

ASTM (American Society for Testing and Materials)	Develops and publishes standard specifications for materials — chemistry, mechanical properties, and test methods.
AWS (American Welding Society)	Develops welding codes and standards, including qualification of welding procedures and welders and filler-metal classification.
ASME (American Society of Mechanical Engineers)	Publishes the Boiler and Pressure Vessel Code (BPVC) and other codes governing the design and fabrication of pressure-critical equipment.
ISO (International Organization for Standardization)	Publishes internationally recognized standards spanning quality systems, materials, and testing, used to harmonize requirements across countries.
API (American Petroleum Institute)	Publishes standards specific to the oil and gas industry, covering pipeline, drilling, and refinery equipment materials and welding.
Indication → Discontinuity → Defect → Reject	The precise evaluation sequence — memorize the distinct legal meaning of each term.
Porosity	A weld, casting, or PM defect consisting of small gas pockets (voids) trapped in the solidified or sintered metal, usually caused by dissolved or entrapped gas that could not escape before solidification finished.
Slag inclusion	A weld defect in which non-metallic slag from the flux becomes trapped within the weld metal or between passes, typically caused by inadequate slag removal between passes and appearing on a radiograph as irregular, elongated dark spots.
Lack of fusion (LOF)	A weld defect in which the weld metal fails to bond completely with the base metal or with a previously deposited pass, leaving an unfused interface, usually caused by insufficient heat input, poor technique, or surface contamination.
Lack of penetration (LOP)	A weld defect in which the weld metal fails to extend fully through the joint root, leaving an unfused gap at the base of the weld, typically caused by insufficient welding current, excessive travel speed, or improper joint preparation.
Undercut / overlap	Weld toe groove vs. rolled-over unfused weld metal — both detectable by VT.
Stress corrosion cracking (SCC) / creep damage	Service-induced defect mechanisms distinct from manufacturing defects.
Non-Conformance Report (NCR)	Formal documentation raised when a component is rejected.
Traceability	Linking an inspection record to the physical component via unique identifiers.
Indication	A response produced by an NDT method — visual, magnetic, ultrasonic, or otherwise — at a specific location on a part; an indication only signals that something was detected and does not by itself mean a rejectable flaw exists.
Discontinuity	An interruption in the normal physical structure or configuration of a part,

	confirmed once an indication has been evaluated; a discontinuity may or may not be severe enough to affect the part's usability.
Defect (NDT/QA evaluation term)	A discontinuity that, when evaluated against a governing code or acceptance criterion, is judged large or severe enough to render the part unfit for its intended use — the finding that triggers a reject decision.
Creep	The slow, time-dependent plastic deformation of a material held under constant stress at elevated temperature, occurring at stress levels well below the yield strength, which can progress to fracture after prolonged service.
Hydrostatic testing	A pressure qualification test in which a vessel or piping system is filled with water and pressurized above its rated working pressure (per code, typically 1.5x) to verify it can safely hold pressure without leaking or failing. It is a required proof test under codes such as ASME and API before a pressure component is put into service.

Unit 4 — Manufacturing Processes

Week 10 — Casting Processes & Defect Recognition

Chill zone	The outermost solidification zone of a casting, made of many small, randomly-oriented grains that nucleate rapidly against the cold mold wall.
Columnar zone	The middle solidification zone of a casting, made of long grains that grow inward from the chill zone toward the center, aligned in the direction of heat flow.
Equiaxed zone	The innermost solidification zone of a casting, made of relatively coarse, randomly-oriented grains that nucleate throughout the remaining liquid once it cools without a strong directional temperature gradient.
Dendrite / dendrite arm spacing (DAS)	Tree-like solidification structure; finer DAS = better mechanical properties.
Riser (feeder)	Reservoir of liquid metal that compensates for solidification shrinkage.
Sand casting	Pattern, mold, cores, gating system — flexible, low tooling cost, rough finish.
Die casting (hot chamber vs. cold chamber)	High-pressure injection into a reusable die — limited to lower-melting alloys.
Investment (lost-wax) casting	Ceramic shell over a wax pattern — best surface finish and accuracy of any casting process.
Shrinkage porosity / gas porosity	Two major casting defect types — irregular voids vs. smooth spherical voids.
Hot tear / cold shut / misrun	Casting defects from mold restraint, cold metal streams not fusing, and incomplete filling.
Gating system	The complete network of channels in a sand-casting mold that carries molten metal from the pouring cup into the mold cavity.
Sprue	The main vertical channel in a sand-casting gating system that receives molten metal from the pouring cup and delivers it downward into the mold.
Runner	The horizontal channel in a gating system that carries molten metal from the sprue and distributes it toward the mold cavity, often to multiple gates.
Gate (casting)	The final, narrow channel in a gating system where molten metal actually enters the mold cavity itself.
Foundry pattern	A reusable, slightly oversized replica of the desired part used to form the mold cavity in sand casting; the oversize allows for solidification and cooling shrinkage.
Mold (casting)	The cavity, typically formed in packed sand or another refractory material around a pattern, into which molten metal is poured and solidifies to take on the shape of the finished casting.
Core (foundry)	A separate sand or ceramic insert placed inside the mold cavity to form

	internal holes, passages, or undercuts that the pattern alone cannot produce.
Die (tooling)	A precision-machined, reusable metal tool containing a cavity or opening that shapes a workpiece under pressure; used to inject metal in die casting and, in other forms, to shape metal in forging, extrusion, and drawing.
Solidification shrinkage	The volumetric contraction a metal undergoes as it transforms from liquid to solid during casting, which must be compensated for with risers or the casting will develop internal voids.

Week 11 — Metal Forming Processes

Recrystallization temperature (Trex)	The boundary (~0.4 Tm) between hot working and cold working.
Hot working vs. cold working	No work hardening vs. work hardening — know the trade-offs of each.
Anisotropy	Directional variation in mechanical properties caused by cold-worked crystallographic texture.
Strain aging	Interstitial atoms pinning dislocations after cold working, raising yield strength (Lüders bands).
Rolling	Reducing a metal's thickness (or forming its cross-section) by passing it between rotating rolls; performed hot or cold and used to produce plate, sheet, and structural shapes.
Hot rolling	Rolling performed above a metal's recrystallization temperature, so the metal continuously recrystallizes as it deforms — no work hardening accumulates, allowing large thickness reductions with lower force but leaving a scaled surface and less precise dimensions.
Cold rolling	Rolling performed below the recrystallization temperature, which work-hardens the metal as it is reduced, giving a smoother surface, tighter dimensional tolerance, and higher strength than hot rolling, but requiring more force and less reduction per pass.
Shape rolling	A rolling process using rolls machined with a matching groove profile to produce long structural shapes with non-flat cross-sections — such as I-beams, channels, angles, and rails — rather than flat plate or sheet.
Forging	A compressive shaping process that hammers or presses metal into shape, producing the highest-integrity wrought parts; performed with open or closed dies.
Open-die forging	Forging between flat or simply-shaped dies that never fully enclose the workpiece, allowing metal to flow outward as it is compressed; used for large, simple shapes and preliminary shaping.
Closed-die (impression-die) forging	Forging between two dies machined with a matching cavity that fully encloses the workpiece, forcing metal to fill the cavity's shape; produces higher-precision, higher-integrity parts with less machining needed afterward.
Grain flow	Elongated, aligned grains following the part's metal-flow direction in forgings — improves fatigue life.
Flash	Excess metal squeezed out at the die parting line in closed-die forging.
Extrusion (direct vs. indirect)	Forcing metal through a die to produce constant-cross-section profiles.
Drawing	Pulling metal through a die (or forming it with a punch) under tension to reduce its cross-section or reshape it — as in wire drawing, tube drawing, and deep drawing.

Wire drawing	Pulling wire stock through a series of progressively smaller conical dies to reduce its diameter and increase its length, work-hardening it in the process.
Deep drawing	A sheet-metal forming process that uses a punch to pull a flat sheet blank into and through a die cavity, forming it into a hollow, cup-like shape without significant change in sheet thickness.
Laps (forming defect)	A forming defect in which metal folds back over itself during rolling or forging without fusing, leaving a thin, sharp discontinuity at the surface.
Seams (forming defect)	A forming defect consisting of a longitudinal surface discontinuity, typically originating from an unhealed surface defect (like a crack) in the original cast or rolled stock that gets elongated during subsequent working.
Bursts (forming defect)	Internal or external cracks that open up during forming when the material's ductility is exceeded at that location, often at the center (centerburst) or surface where tensile stresses are highest.
Pipe defect	A funnel-shaped shrinkage cavity that forms at the top-center of an ingot or billet during solidification, which — if not cropped off before working — gets drawn out into a long internal void running through the worked product.
Chevron (center-burst) cracking	A V-shaped internal crack pattern that forms along the centerline of drawn or extruded rod/wire/tube when excessive hydrostatic tension in the deformation zone pulls the material apart from the inside.
Crystallographic texture	A preferred, non-random crystallographic orientation of grains in a polycrystalline metal, typically developed during cold working, that causes mechanical properties to vary with direction.
Lüders bands	Visible bands of localized plastic deformation that sweep across a metal's surface at the onset of yielding, producing a rough, stretcher-strain surface pattern; common in strain-aged or low-carbon steels.
Parting line	The line where the two halves of a die or mold meet; excess material (flash) is squeezed out and forms along this line during closed-die forging.
Punch (tooling)	A hardened tool that pushes or pulls metal into or through a die opening, used alongside or instead of a die in operations such as drawing, stamping, and deep drawing.
Draft angle	The slight taper designed into the vertical walls of a die or mold cavity so the formed part can be withdrawn cleanly from the tooling without dragging, galling, or being damaged on ejection.
Billet	A short, thick, solid section of metal — usually cast or rolled to a round or square cross-section — that serves as the starting stock fed into a forging or extrusion process.
Plane strain	A loading condition in which deformation is essentially zero along one direction (typically the width, held fixed by surrounding material) while the material strains freely in the other two directions; it defines the most severe, lowest-ductility point on a forming limit curve.
Wrinkling (sheet metal forming defect)	A sheet-forming defect in which the sheet buckles into waves rather than stretching smoothly, occurring under biaxial compressive strain states (as opposed to necking or fracture, which occur under stretching strain states).

Tube drawing methods (sinking, fixed plug, floating plug, mandrel)	Four variations of tube drawing that differ in how the tube's inside diameter is controlled while it is pulled through a reducing die: sinking uses no internal tool at all, a fixed plug is held in place on a rod, a floating plug rides freely inside the tube, and a mandrel is a long rod supporting the full length of the bore.
Hydrostatic tension	A state of stress in which the deformation zone experiences tension in all three directions simultaneously; excessive hydrostatic tension during drawing pulls internal voids open, producing central-burst (chevron) cracking along the tube or wire's centerline.

Week 12 — Machining

Shear plane / primary, secondary, tertiary deformation zones	Where chip formation, tool-chip friction, and flank rubbing occur during cutting.
Continuous vs. discontinuous chip	Chip type depending on material ductility and cutting parameters.
Built-up edge (BUE)	Workpiece material welding to the tool at low cutting speed, causing a rough finish.
Cutting speed (Vc)	The speed at which the cutting edge moves relative to the workpiece surface, usually in meters per minute (m/min); set primarily by the tool material and the workpiece being cut.
Feed (f)	The distance the cutting tool advances relative to the workpiece per revolution (turning) or per tooth (milling), which — together with the tool's nose radius — largely determines the resulting surface finish.
Depth of cut (ap)	The thickness of the layer of material removed in a single pass, measured perpendicular to the machined surface.
Turning	A machining process in which a cutting tool removes material from a rotating workpiece on a lathe to produce cylindrical or conical external and internal surfaces.
Milling	A machining process in which a rotating multi-tooth cutter removes material from a stationary or moving workpiece to produce flat surfaces, slots, pockets, and contoured features.
Drilling	A machining process in which a rotating drill bit is fed axially into a workpiece to produce a round hole.
Grinding (machining)	A machining/finishing process that uses a rotating abrasive wheel to remove very thin layers of material, producing fine surface finishes and tight dimensional tolerances, often as a final hardening/finishing step.
Up milling vs. down (climb) milling	Chip-thickness direction and resulting surface finish/residual stress differences.
HSS (High-Speed Steel)	A tough, relatively inexpensive tool-steel material that retains hardness at moderately elevated temperatures; used for general-purpose cutting tools, especially where toughness or cost matters more than maximum cutting speed.
Carbide (cemented/tungsten carbide)	A hard, wear- and heat-resistant tool material made from tungsten carbide particles bonded in a cobalt matrix; runs at much higher cutting speeds than HSS and is the standard modern cutting-tool material.
Ceramic (cutting tool)	An extremely hard, heat-resistant but brittle tool material (e.g., aluminum-oxide or silicon-nitride based) used for very high-speed finishing cuts on hard materials, where the setup is rigid enough to avoid chipping the brittle edge.
CBN (Cubic Boron Nitride)	The second-hardest known material after diamond; used to machine hardened steels and superalloys at high speed, where diamond tooling would

	chemically react with the iron in the workpiece.
PCD (Polycrystalline Diamond)	A diamond-based cutting tool material offering extreme hardness and wear resistance for machining non-ferrous metals and composites (not used on steel, which reacts chemically with diamond at cutting temperatures).
Surface roughness: Ra and Rz	Arithmetic mean roughness vs. peak-to-valley height — the two key surface finish parameters.
Grinding burn (re-tempering vs. re-hardening burn)	Thermal damage from excessive grinding heat — detected by Nital etch.
Chatter marks	Vibration-induced wave pattern on a machined surface.
Recast layer (EDM)	Re-solidified surface layer from electrical discharge machining containing tensile residual stress.
Tool flank / flank wear	The flank is the tool face that rubs against the freshly cut workpiece surface behind the cutting edge; flank wear is the gradual abrasive erosion of that face and is a primary measure used to judge tool life.
Nital etch	A dilute nitric acid-in-alcohol etching solution applied to a ground steel surface to reveal thermal damage such as grinding burn, staining re-hardened and re-tempered zones differently from the unaffected base metal.
Nose radius (tool tip radius)	The small radius ground onto the tip of a cutting tool insert ($r\varepsilon$), rather than a sharp point. It directly sets the theoretical surface roughness in turning ($Ra \approx f^2/8r\varepsilon$) — a larger nose radius at the same feed produces a smoother finish.
Uncut chip thickness (t_1)	The thickness of the layer of material being removed ahead of the cutting edge, before it deforms into the actual chip. It is set by the feed and depth of cut and is used with the width of cut to compute the shear-plane area and specific cutting energy.
Included angle (cutting tool)	The angle formed at the point of a single-point cutting tool (or drill) between its two converging cutting edges; it governs tool tip strength and how the tool shears the chip during machining.
Countersink	A conical recess machined at the mouth of a hole so a flat-head fastener seats flush with, or below, the surface; specified by its half-angle (commonly 41° or 45°) to match the fastener head's taper.
Keyway	A machined slot cut into a shaft and mating hub to seat a key, which transmits torque between the two; the sharp internal corner of a keyway is a common stress-concentration site and fatigue crack-initiation point.
Thread root	The bottom of the V-shaped groove cut into a fastener or threaded shaft; because it is the smallest cross-section along the thread and has a sharp geometric transition, it is typically the highest-stress, most fatigue-critical location on a threaded part.
Rake angle	The angle between a cutting tool's top face and a line perpendicular to the workpiece surface; a positive rake angle reduces cutting forces and produces a shearing action, while a negative rake angle strengthens the edge for hard or interrupted cuts.

Size effect (machining)	The tendency for specific cutting energy (energy per unit volume of material removed) to increase as the uncut chip thickness gets smaller, because a larger fraction of the cutting energy goes into plowing and friction rather than shearing as the chip gets thinner.
Thrust force (Ft)	The force component in machining acting perpendicular to the cutting velocity, pushing the tool away from or into the workpiece; distinct from the cutting force (Fc) that acts along the direction of tool travel, it can even reverse sign as the rake angle and depth of cut change.
Scallop height	The tiny cusps of uncut material left between adjacent passes of a ball-nose end mill, caused by the tool's rounded tip not fully overlapping the previous pass; it is the dominant factor controlling surface roughness in ball-end milling.
Chip load	The thickness of material each cutting edge removes per revolution (feed per tooth); it must be increased for a given cut when the effective engagement is a small arc of the cutter, to avoid rubbing instead of cutting (chip thinning).

Week 13 — Powder Metallurgy

Water atomization vs. gas atomization	Powder production methods producing irregular vs. spherical particle shapes.
Green compact / green strength	The fragile, pressed-but-unsintered powder part before sintering.
Sintering	Densification by solid-state atomic diffusion below the melting point — not melting.
Liquid phase sintering	A lower-melting additive melts and fills pores by capillary action (key to WC-Co tools).
Hot isostatic pressing (HIP)	Simultaneous heat + isostatic gas pressure — eliminates porosity, approaches wrought properties.
Cold isostatic pressing (CIP)	Room-temperature isostatic pressing — eliminates density gradients in complex shapes.
Metal injection molding (MIM)	Fine powder + polymer binder injection molded like plastic, then debound and sintered.
Density gradient	A PM defect in which density varies from one region of a pressed part to another, usually caused by uneven powder fill or uneven pressure transmission during compaction.
Lamination cracks	A PM defect consisting of planar cracks running perpendicular to the pressing direction, typically caused by trapped air or spring-back as the green compact is ejected from the die.
Blistering	A PM defect in which surface bubbles or bulges form during sintering, typically caused by burning off the lubricant/binder too quickly, trapping gas beneath the surface.
Additive manufacturing	A family of layer-by-layer metal powder or wire fusion processes that build a part directly from a 3D model, with no tooling required.
SLM (Selective Laser Melting)	A metal additive manufacturing process that uses a laser to fully melt and fuse successive layers of metal powder in a powder bed, building a solid part directly from a 3D model with no tooling.
EBM (Electron Beam Melting)	A metal additive manufacturing process similar to SLM but using a focused electron beam instead of a laser, performed in a vacuum, which allows faster builds and preheats the powder bed to reduce residual stress.
DED (Directed Energy Deposition)	A metal additive manufacturing process that feeds powder or wire through a nozzle into a focused energy source (laser, electron beam, or arc) that melts it on deposition, allowing material to be added onto an existing part rather than only building in a powder bed.
Wrought (metal)	Metal that has been shaped by mechanical deformation processes such as rolling, forging, extruding, or drawing, giving it a worked grain structure with mechanical properties generally superior to as-cast metal.
Debinding	The step in metal injection molding where the polymer binder is removed from the molded green part, by thermal or solvent means, leaving a fragile

	powder skeleton ready for sintering.

Unit 5 — Joining, Failure & Corrosion

Week 14 — Welding Processes

SMAW (Shielded Metal Arc Welding / Stick)	Coated consumable electrode; flux forms shielding gas and slag.
GMAW (Gas Metal Arc Welding / MIG)	Continuous bare wire with shielding gas; transfer modes: short circuit, spray, pulsed.
GTAW (Gas Tungsten Arc Welding / TIG)	Non-consumable tungsten electrode; highest quality; AC for aluminum, DCEN for steel.
FCAW (Flux-Cored Arc Welding)	Tubular flux-filled wire; self-shielded (FCAW-S) vs. gas-shielded (FCAW-G).
Heat input formula	$HI = (V \times A \times 60) / (1000 \times \text{travel speed}) \rightarrow$ kJ/mm — controls cooling rate and grain size.
AWS filler metal classification (e.g., E7018, ER308L, ER4043)	Decode prefix, strength, position, and flux/chemistry codes.
Preheat / interpass temperature	Minimum temperature before/during welding to slow cooling and prevent HAZ cracking.
Carbon equivalent (CE) and preheat thresholds	CE determines required preheat — memorize the CE bands (<0.40, 0.40–0.60, >0.60).
Weld symbol anatomy (reference line, arrow/other side, tail)	How to read the AWS A2.4 weld symbol standard.
Butt joint	A joint in which two pieces are aligned edge-to-edge in the same plane and welded along the abutting edges.
T-joint	A joint in which one piece meets another at a right angle, forming a T-shape, welded along the intersection (typically as a fillet weld).
Lap joint	A joint in which two overlapping pieces are welded along the edge of the overlap.
Corner joint	A joint in which two pieces meet at a corner (typically near 90°) and are welded along the intersection, commonly forming an L-shape.
Edge joint	A joint in which two pieces lie parallel and roughly flush with each other and are welded along their common edge.
Square groove	A butt-joint edge preparation with no bevel — the mating edges are left square (flat), used only for thin material that a single pass can fully penetrate.
V-groove	A butt-joint edge preparation in which each mating edge is beveled to form a V-shaped gap, used on thicker material to allow full penetration in one or a few passes.

U-groove	A butt-joint edge preparation with a rounded, U-shaped gap machined into thick material, reducing the amount of filler metal needed compared to a V-groove of the same thickness.
Bevel groove	An edge preparation in which only one of the two mating pieces is angled (beveled), commonly used at a T-joint or corner joint rather than a straight butt joint.
Electrode (welding)	The component that carries welding current into the arc; it may be consumable, melting to supply filler metal (as in SMAW, GMAW, FCAW), or non-consumable, such as the tungsten electrode in GTAW, which sustains the arc without melting.
Flux (welding)	A chemical coating or core material on a welding electrode/wire that shields the molten weld pool from atmospheric oxygen and nitrogen, cleans the metal, and decomposes into shielding gas and a protective slag layer.
Slag	The solid, glassy layer of spent flux that forms over a cooling weld bead; it protects the solidifying metal from the atmosphere and must be chipped or brushed off before the next pass or inspection.
Shielding gas	An inert or semi-inert gas (e.g., argon, CO2, or argon-CO2 blends) fed around the arc and weld pool to exclude atmospheric oxygen and nitrogen, preventing porosity, oxidation, and nitride formation in the weld.
Metal transfer modes (short circuit / spray / pulsed)	The ways molten filler droplets move from the GMAW wire to the weld pool: short circuit (low heat, droplets bridge the gap repeatedly), spray (high current, fine continuous droplet stream), and pulsed (current pulses release one controlled droplet per pulse).
DCEN (Direct Current Electrode Negative)	A welding current polarity in which the electrode is connected to the negative terminal and the workpiece to the positive; electron flow concentrates heat in the workpiece, giving deeper penetration, commonly used for GTAW on steel.
Welding position	The orientation of the joint and weld axis during welding — flat, horizontal, vertical, or overhead — which governs technique, weld-pool control, and is encoded in filler metal and electrode classification numbers.
Travel speed (welding)	The rate at which the welding arc moves along the joint, usually given in mm/min or in/min. It sits in the denominator of the heat input formula, so a slower travel speed at the same voltage and amperage delivers more heat input and a slower cooling rate.
Fillet weld	A weld that joins two pieces meeting at roughly a right angle (e.g., a T-joint or lap joint) by depositing weld metal into the corner, producing a triangular cross-section rather than penetrating fully through the joint.
Leg size (weld)	The distance from the weld root along one of the original base-metal surfaces out to the toe of the weld; for a standard equal-leg fillet weld both legs are the same length, denoted w.
Weld root / Weld face	The weld root is the point where the two base-metal surfaces meet at the back/bottom of the joint, marking the deepest extent of fusion; the weld face is the exposed outer surface of the finished weld bead on the side it was deposited.

SAW (Submerged Arc Welding)	An arc welding process in which a continuously fed bare wire electrode is shielded not by gas but by a blanket of granular flux poured ahead of the arc; it produces very high deposition rates and deep penetration, commonly used for long, straight, thick-section welds.

Week 15 — Weld Metallurgy & Failure

Heat-affected zone (HAZ)	Base metal heated above ~730°C but not melted — the most failure-prone weld region.
CGHAZ / FGHAZ / ICHAZ / SCHAZ	The four HAZ sub-zones by peak temperature — coarse-grained, fine-grained, intercritical, subcritical.
Acicular ferrite / allotriomorphic ferrite	Weld metal microstructures — the desirable fine interlocking plates vs. the crack-prone grain-boundary phase.
Dilution	The proportion of melted base metal mixed into the deposited weld metal.
Hot cracking (solidification cracking)	Cracking from low-melting films at grain boundaries during solidification; controlled by Mn:S ratio.
Hydrogen-induced cracking (HIC) / cold cracking	Delayed cracking requiring three simultaneous conditions: susceptible microstructure, hydrogen, tensile stress.
Lamellar tearing	Through-thickness base-metal cracking from elongated MnS inclusions under weld shrinkage stress; prevented with Z-quality steel.
Ductile fracture (microvoid coalescence, cup-and-cone)	Fracture with plastic deformation, dimpled surface.
Brittle fracture (cleavage, chevron marks, river marks)	Sudden fracture with no plastic deformation; flat, crystalline surface.
Fatigue fracture (beach marks, striations, ratchet marks)	Progressive fracture from cyclic loading; beach marks visible to the eye, striations only under SEM.
Failure analysis methodology	Preserve → document → examine → NDT → mechanical/metallographic test → root cause → report.
Mn:S ratio	The ratio of manganese to sulfur content in steel; adequate manganese ties up sulfur as manganese sulfide instead of low-melting iron sulfide films at grain boundaries, reducing susceptibility to solidification (hot) cracking.
Z-quality steel	Steel that has been tested and certified for through-thickness ductility (e.g., per EN 10164), achieved through low sulfur content and controlled inclusion shape/size, used to resist lamellar tearing.
SEM (Scanning Electron Microscope)	An electron microscope that scans a focused beam of electrons across a specimen surface to produce high-magnification, high-depth-of-field images, used in fractography to resolve fine fracture features such as fatigue striations.
NDT (Non-Destructive Testing)	Inspection methods — visual, penetrant, magnetic particle, ultrasonic, radiographic, and eddy current testing — used to detect defects in a part or weld without damaging or altering it.
Metallography	The preparation (sectioning, mounting, grinding, polishing, and etching) and microscopic examination of a material's surface to reveal and characterize

	its internal microstructure.
Stress intensity factor (ΔK)	A fracture-mechanics parameter that describes the magnitude of the stress field at a crack tip; ΔK is the range of this value over one loading cycle (K_max - K_min). It drives fatigue crack growth rate in Paris' Law, da/dN = C(ΔK)^m, and should not be confused with the stress concentration factor Kt.
FEA (Finite Element Analysis)	A computer-based numerical method that divides a complex part geometry into small elements to calculate stress, strain, and deflection under load; commonly used to find stress concentration factors (Kt) for geometries too complex for standard charts.
Shot peening	A cold-working process that bombards a metal surface with small round media (shot) at high velocity, plastically stretching the surface layer and leaving it in residual compression, which delays fatigue crack initiation.
TIG dressing	A post-weld improvement technique in which a GTAW (TIG) torch, without filler metal, remelts and reshapes the weld toe to remove undercut and reduce the sharp geometric transition there, lowering the local stress concentration and improving fatigue life.
Stage II fatigue crack growth	The middle, stable phase of fatigue crack growth, following initiation (Stage I) and preceding rapid final fracture (Stage III), during which the crack advances a small, predictable increment with each load cycle — the phase described by Paris' Law.
Ferrite Number (FN)	A standardized index (not a percentage) that quantifies the amount of ferrite present in an austenitic or duplex stainless steel weld deposit; a target FN of roughly 4-8 balances enough ferrite to resist solidification (hot) cracking against too much, which risks embrittlement.

Week 16 — Corrosion & Course Wrap-up

Four-element corrosion cell (anode, cathode, electrolyte, metallic path)	All four must be present for electrochemical corrosion — remove one to stop it.
Galvanic series	Ranking of metals by electrochemical potential — determines which metal corrodes when dissimilar metals contact.
Passivation	Formation of a stable, adherent oxide layer that slows corrosion (stainless, aluminum, titanium).
Uniform (general) corrosion	Even, predictable surface loss measured in mm/year or mils per year (MPY).
Galvanic corrosion / area effect	Accelerated corrosion of the less-noble metal when coupled to a more-noble one; area ratio matters.
Pitting corrosion / PREN	Localized passive-film breakdown; Pitting Resistance Equivalent Number quantifies alloy resistance.
Crevice corrosion	Localized attack in stagnant, oxygen-depleted gaps (gaskets, lap joints).
Stress corrosion cracking (SCC)	Tensile stress + specific corrosive environment causes cracking below yield strength.
Cathodic protection (sacrificial anode vs. impressed current)	Makes the structure the cathode so corrosion occurs only at a sacrificial anode.
Hot-dip galvanizing / thermal spray / anodizing	Common protective coating methods — know the mechanism of each.
Corrosion under insulation (CUI)	Hidden corrosion beneath insulation, detected by pulsed eddy current or radiographic profile.
Anode	The site in a corrosion cell where oxidation occurs and metal atoms give up electrons to become ions; the anode is where metal loss (corrosion damage) actually takes place.
Cathode	The site in a corrosion cell where reduction occurs (such as oxygen reduction or hydrogen evolution); the cathode is protected from metal loss while the coupled anode corrodes.
Electrolyte	A conductive liquid, such as moisture, water, or soil solution, that allows ion flow between the anode and cathode, completing the electrical circuit required for electrochemical corrosion.
Electrochemical potential	A measured value indicating a metal's tendency to give up electrons (oxidize) relative to a reference electrode; more negative potentials mark more active (anodic) metals and more positive potentials mark more noble (cathodic) metals.
Impressed current	A cathodic protection method that uses an external DC power source to drive current through inert anodes to the structure, forcing the structure to behave as the cathode instead of relying on a sacrificial metal.
Hot-dip galvanizing	A coating process in which steel is immersed in a bath of molten zinc,

	forming a metallurgically bonded zinc and zinc-iron alloy layer that protects the steel both as a physical barrier and sacrificially.
Thermal spray coating	A coating process in which a metal, ceramic, or alloy feedstock is heated and propelled as molten or semi-molten droplets onto a prepared substrate, building up a mechanically bonded protective layer.
Anodizing	An electrochemical process, typically applied to aluminum, in which the part serves as the anode in an electrolytic cell so that its natural oxide layer thickens and hardens, improving corrosion and wear resistance.
Corrosion allowance	Extra material thickness added to a component's design, beyond what is needed to carry the load, specifically to accommodate expected material loss from corrosion over the part's intended service life.
Reference electrode ($Cu/CuSO_4$)	A copper rod immersed in a saturated copper sulfate solution, used as a stable, known-potential reference point against which the protection potential of a cathodically protected structure is measured in the field (e.g., the -0.85 V criterion).

Reference Material — Designation Systems & Field Skills

Steel, Aluminum, Copper & Titanium Designations

AISI/SAE steel series	A four-digit prefix system identifying a steel's alloy family — the first two digits denote the alloying family and the last two denote carbon content in hundredths of a percent.
AISI/SAE 10xx series	The plain carbon steel series — no significant alloying elements beyond carbon and manganese.
AISI/SAE 41xx series	The chromium-molybdenum ("chromoly") alloy steel series — good hardenability and toughness, common in aerospace tubing and fasteners.
AISI/SAE 43xx series	The nickel-chromium-molybdenum alloy steel series — high strength and toughness, used in highly stressed parts like gears and aircraft components.
AISI/SAE 51xx series	The chromium alloy steel series — good hardenability, commonly used for springs and bearings.
AISI/SAE 52100	A high-carbon chromium alloy steel formulated specifically for bearing applications, valued for its ability to be hardened uniformly and resist rolling-contact fatigue.
Aluminum Association temper codes	The letter-and-number suffix system after an aluminum alloy number that specifies the alloy's actual heat-treatment/work-hardening condition — and therefore its real strength and hardness.
Temper O (aluminum)	The fully annealed condition — the softest, most ductile, lowest-strength temper for a given aluminum alloy.
Temper H (aluminum)	The strain-hardened (cold-worked) condition; a following digit indicates the degree of hardening, with no thermal aging involved.
Temper T4 (aluminum)	Solution heat-treated and naturally aged at room temperature to a stable condition.
Temper T6 (aluminum)	Solution heat-treated and artificially aged (aged at an elevated temperature) — the most common heat-treated condition for structural aluminum alloys like 6061 and 7075.
Temper T651 (aluminum)	A T6 temper that has additionally been stress-relieved by stretching after solution treatment, used for plate that must machine flat without distortion.
UNS copper numbering (Cxxxxx)	Unified Numbering System blocks that classify coppers, brasses, bronzes, and copper-nickels by composition.
CDA (Copper Development Association)	The copper industry trade association that developed and maintains the wrought and cast copper alloy numbering system (Cxxxxx), which forms the basis for the UNS designations covering coppers, brasses, bronzes, and copper-nickels.
Copper-nickel alloys	Copper alloys in which nickel (roughly 5-30%) is the principal alloying element, forming a continuous solid solution with copper; valued for outstanding resistance to seawater corrosion and biofouling, so they are widely used in marine piping and condenser tubing.

ASTM titanium grades	Grade number carries no composition meaning by itself — titanium grades are sorted into CP/alpha, near-alpha, alpha-beta, and beta families based on actual alloy chemistry (Grade 2 is CP titanium, covered under CP Titanium above).
ASTM Grade 5 titanium (Ti-6Al-4V)	The most widely used titanium alloy — an alpha-beta alloy with 6% aluminum and 4% vanadium, offering a strong combination of strength, weldability, and corrosion resistance.
ASTM Grade 23 titanium (Ti-6Al-4V ELI)	An Extra-Low-Interstitial version of Ti-6Al-4V with tighter oxygen/interstitial control, giving higher ductility and fracture toughness for critical applications like biomedical implants.
Extra Low Interstitial (ELI)	Titanium grade variant with tighter oxygen/interstitial control for higher ductility and fracture toughness.

Quality Codes, PPE & Shop Practice

AWS D1.1 (Structural Welding Code — Steel)	Governing code for welded steel structures; know its major clause structure.
ASME BPVC (Boiler and Pressure Vessel Code)	Governs pressure-critical equipment design and fabrication.
Welding Procedure Specification (WPS)	Document specifying process, filler metal, preheat, and parameters for a qualified weld.
Prequalified joint / procedure qualification	Joint details that don't require separate procedure testing under a code.
ANSI Z87.1 / Z87+	Eye protection rating standards for basic vs. high-impact hazards.
Sensitization vs. passivation (shop context)	Reinforce these terms in the context of fabrication practice, not just theory.

Units of Measurement & Dimensional Analysis

The SI Base Units

Base unit (concept)	One of seven independently defined units from which every other physical quantity's unit is built by multiplication and division. None of the seven can be expressed in terms of the others.
Length — meter (m)	The base unit for size, distance, diameter, thickness, and depth. This course mixes it constantly with millimeters (mm) and, in U.S. shop practice, inches (in).
Mass — kilogram (kg)	The base unit for the amount of matter in an object — distinct from weight (a force). Alloy compositions are reported in mass % (wt%), not the mole-based at% used in some materials science contexts.
Time — second (s)	The base unit underlying every rate: cooling rate (°C/s), cutting speed (m/min), frequency (Hz = 1/s), corrosion rate (mm/year).
Temperature — kelvin (K)	The SI base unit; this course reports working temperatures in Celsius (°C) almost exclusively, with Fahrenheit (°F) appearing in AWS D1.1 preheat tables and some U.S. industry references.
Electric current — ampere (A)	The base unit for current — shows up directly in welding (arc amperage) and in the heat input formula.
Amount of substance — mole (mol)	Rarely used directly in this course; alloy chemistry is reported by mass percent instead, but this is the base unit behind it in formal materials science.
Luminous intensity — candela (cd)	The base unit behind illuminance (lux) — relevant to the minimum lighting requirement for visual testing (VT ≥ 500 lux).
Why only 4 of the 7 dominate this course	Nearly every equation in this glossary is built from just length, mass, time, and temperature. Recognizing that shrinks the whole unit system down to a handful of building blocks.

Dimensional Analysis — Building Derived Units

Derived unit (concept)	A unit for a quantity that is not one of the seven base quantities — built by combining base units the same way the physical formula combines the base quantities. If $\sigma = F/A$, then the unit for σ must be (unit of F)/(unit of A).
Force — newton: N = kg·m/s²	From Newton's second law, $F = m \cdot a$. Mass (kg) times acceleration (m/s^2). Every load in this course — cutting force Fc, hardness test load, Charpy hammer weight — is ultimately a force in newtons (or kgf, see below).
Stress / Pressure — pascal: Pa = N/m² = kg/(m·s²)	From $\sigma = F/A$ — force divided by area. This is the derived unit behind every stress and pressure value in the course: yield strength, UTS, PREN's implicit chloride environment, HIP pressure, hydrostatic test pressure — all reported in Pa, kPa, MPa, or GPa.
Energy / Work — joule: J = N·m = kg·m²/s²	From $W = F \cdot d$ (force times distance) or $E = mgh$ (the Charpy/Izod energy formula). This is the unit behind impact toughness, and — divided by volume — behind specific cutting energy (J/mm^3).
Power — watt: W = J/s = kg·m²/s³	Energy per unit time. This is the unit behind cutting power $Pc = Fc \cdot Vc/(60{,}000 \cdot \eta)$ and behind electrical power in welding (V × A).
Density — kg/m³ (or g/cm³)	Mass per unit volume — directly built from the two base units mass and length³. Used to compare families at a glance: aluminum ≈ 2.7 g/cm^3, titanium ≈ 4.5 g/cm^3, steel ≈ 7.85 g/cm^3.
Frequency — hertz: Hz = 1/s	Cycles per second — the unit behind UT/ET sound and induction frequency (1–10 MHz for UT; 1 kHz–1 MHz for ET) and behind fatigue loading rate.
The general method	Write the defining equation for the quantity → substitute the base (or already-derived) units for each variable → simplify algebraically. The result is guaranteed correct as long as the defining equation is correct — this is why dimensional analysis works as a check on any formula, not just a way to build new ones.

Worked Examples — Checking a Formula's Units

Heat input: HI = (V·A·60) / (1000 · travel speed)	V·A = watts = J/s. Multiplying by 60 (s/min) converts to J/min, matching the travel speed's per-minute basis. Dividing by (mm/min) leaves J/mm; dividing by 1000 converts J to kJ — confirming the stated result of kJ/mm.
Turning roughness: $Ra = f^2 / (8r)$	f (feed, mm) squared divided by r (nose radius, mm) leaves units of mm — matching Ra's reported units (usually converted to µm by ×1000). If f or r were entered in the wrong unit, the result would be off by orders of magnitude — exactly why unit consistency is checked before trusting a calculated Ra.
Specific cutting energy: $u = Fc / (t_1 \cdot b)$	Fc in newtons, t_1 and b in mm — so $t_1 \cdot b$ is an area (mm^2). N/mm^2 = MPa, but specific cutting energy is conventionally reported as J/mm^3 once cutting distance is folded in — the units track exactly with the physical picture of 'energy consumed per unit volume of metal removed.'
Why hardness numbers are the exception	HRC, HB, and HV are empirical index numbers, not true derived SI units — HB and HV do reduce to a load-over-area (kgf/mm^2) inside the formula, but the numbers are calibrated against standardized indenters and loads, so a '60 HRC' cannot be dimensionally converted to '60 HB' the way MPa converts to psi. Cross-scale conversion requires an empirical lookup table, not algebra.
Dimensionless numbers	Some of the most important quantities in this course carry no units at all: carbon equivalent (CE), PREN, strain (ε), the strain-hardening exponent (n), notch sensitivity (q), and stress concentration factor (Kt) are all ratios of like quantities — their units cancel completely, which is exactly why they can be used to rank or compare materials regardless of specimen size.

Shop-Floor & U.S. Customary Units Still in Use

ksi / psi (stress, pressure)	1 ksi = 1000 psi = 6.895 MPa. Used constantly in AWS filler classification (E7018 = 70 ksi minimum tensile) and ASME/API pressure ratings alongside SI values.
ft·lbf (energy)	1 ft·lbf = 1.356 J. The U.S. customary unit for Charpy/Izod impact energy, reported alongside or instead of joules.
°F (temperature)	°C = (°F − 32) / 1.8. AWS D1.1 preheat/interpass tables (Table 4.4) list both °F and °C side by side — always confirm which one a spec is using before setting a preheat torch or oven.
in / mil (length)	1 in = 25.4 mm exactly. 1 mil = 0.001 in = 0.0254 mm — the standard unit for coating thickness and for corrosion rate (mils per year, MPY).
kgf (force / load)	1 kgf = 9.81 N. Hardness test loads (Brinell, Vickers) are conventionally specified in kgf even though the underlying physics is a force in newtons.
lux (illuminance)	The SI-derived unit for VT lighting requirements (≥ 500 lux, or per procedure) — built from candela and area, rarely discussed but directly specified in inspection procedures.
Rule of thumb for this course	When a number looks unfamiliar, check its unit before its magnitude — a '150' could be °F (preheat), HB (hardness), or MPa (a low stress) depending entirely on the unit attached.

Quick Math Review — Skills for Using These Equations

Order of Operations

PEMDAS	Parentheses → Exponents → Multiplication & Division (left to right) → Addition & Subtraction (left to right). This order is not optional — evaluating a formula in the wrong sequence gives a wrong number that still looks plausible.
Worked example: CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15	Each division happens first and only applies to its own term — Mn/6 divides Mn alone, not the running total. Compute all four terms separately, then add them last.
Common mistake: dividing the whole sum instead of one term	Writing (C + Mn)/6 instead of C + Mn/6 changes the answer completely — watch for which quantities are actually inside the denominator's 'reach.'
Grouping symbols change the order	In $HB = 2F / [\pi D(D - \sqrt{(D^2 - d^2)})]$, everything inside the brackets must be fully resolved — including the square root — before dividing 2F by it.
Nested operations, innermost first	Work from the innermost parentheses outward: in $D - \sqrt{(D^2 - d^2)}$, compute D^2 and d^2 first, subtract, then take the square root, then subtract from D.

Fractions & Ratios

A fraction is a ratio	a/b compares a to b. Stress $\sigma = F/A$ is literally 'force per unit area' — the fraction bar means 'for each unit of the denominator.'
Fractions with different units are still valid ratios	Fatigue ratio = $S_e \div$ UTS divides a stress by a stress, so the units cancel and the result is a pure (dimensionless) number — useful for comparing materials regardless of size.
Cross-multiplication solves a proportion	If $a/b = c/d$, then $a \cdot d = b \cdot c$. This is exactly how a unit conversion works: 1 in / 25.4 mm = x in / 100 mm → x = 100/25.4.
Multiplying by a conversion factor of 1	Any conversion factor (e.g., 25.4 mm/1 in) equals 1 in value, so multiplying by it changes only the units, never the physical quantity — this is the arithmetic behind every unit conversion in the previous section.
Reciprocals flip a ratio	If $r = t_1/t_2$ (chip thickness ratio), then $t_2/t_1 = 1/r$ — useful when a formula or a data table reports the inverse of the ratio you need.

Percentages

Percent means 'per hundred'	37% = 37/100 = 0.37. Convert to decimal before using in most formulas unless the formula is explicitly written to accept the number as-is (see below).
Watch which formulas want a whole number vs. a decimal	PREN and CE use %Cr, %Mn, etc. as the whole number (e.g., 18 for '18% Cr'), not 0.18 — plug in exactly what the alloy spec lists. Strain as a percentage, by contrast, is usually converted to a decimal (2% strain = 0.02) before use in stress-strain equations.
Percent change formula	% change = (final − initial)/initial × 100%. This single pattern generates %EL (elongation), %RA (reduction in area), and %CW (percent cold work) — only the quantity being compared changes.
AISI/SAE carbon 'points' are hundredths of a percent, not whole percent	The last two digits of a steel grade (e.g., the '45' in 1045) mean 0.45% carbon — misreading this as '45% carbon' is a factor-of-100 error with a very different (impossible) steel.
Percentages don't simply add across different bases	10% of one alloy's Cr plus 5% of another's Mo cannot be added directly unless both percentages are of the same total mass — always check what the percentage is 'of' before combining values from different sources.

Exponents & Powers

Basic exponent rules	$x^a \cdot x^b = x^{a+b}$ (multiply, add exponents). $(x^a)^b = x^{ab}$ (power of a power, multiply exponents). $x^{-1} = 1/x$ (negative exponent means reciprocal).
Squares in area and geometry	d^2 and D^2 in the Brinell formula are just the diameters multiplied by themselves — always compute the square before combining with other terms per order of operations.
Fractional exponents in the Hollomon equation: $\sigma = K \cdot \varepsilon^n$	n is typically a decimal between 0 and 1 (e.g., 0.15–0.25 for steel). Raising strain to a fractional power means flow stress rises more slowly than strain itself — a bigger n means the material keeps hardening (and resisting necking) longer.
Solving for an exponentiated variable: Taylor's tool life, $Vc \cdot T^n = C$	Isolate $T^n = C/Vc$, then undo the exponent by raising both sides to the power 1/n: $T = (C/Vc)^{(1/n)}$. This 'undo with the reciprocal power' move works for any equation of the form $x^n =$ (something).
Powers of a stress-intensity range: Paris' Law, $da/dN = C(\Delta K)^m$	ΔK is raised to a power m before multiplying by the constant C — a small change in ΔK produces a much larger change in crack growth rate whenever $m > 1$, which is typical (m is often 2–4 for metals).

Square Roots & Radicals

A square root undoes a square	$\sqrt{x}$ is the number that, multiplied by itself, gives x. Equivalently, $\sqrt{x} = x^{(1/2)}$ — a square root is just an exponent of one-half.
Radical of a product splits apart: $\sqrt{(a \cdot b)} = \sqrt{a} \cdot \sqrt{b}$	Used to read the skin depth formula $\delta = 503/\sqrt{(f \cdot \mu_r \cdot \sigma)}$ — you can evaluate $f \cdot \mu_r \cdot \sigma$ first and take one square root, or take three separate square roots and multiply; both give the same answer.
Solving for a variable trapped under a root: Hall-Petch, $\sigma_\gamma = \sigma_0 + k/\sqrt{d}$	Isolate the root term: $k/\sqrt{d} = \sigma_\gamma - \sigma_0$. Take the reciprocal of both sides, then square: $d = [k/(\sigma_\gamma - \sigma_0)]^2$. Squaring is how you 'undo' a square root once it's alone on one side.
Order matters inside a radical	In $D - \sqrt{(D^2 - d^2)}$, the subtraction $D^2 - d^2$ must be completed before the square root is taken — $\sqrt{(D^2)} - \sqrt{(d^2)}$ is NOT the same thing and gives a wrong (and much larger) answer.
A negative number under an even root has no real answer	If a calculation under a square root ever comes out negative, that's a signal a value was plugged in wrong (e.g., d entered larger than D in the Brinell formula) — it is not a sign to just drop the negative sign.

Scientific Notation & Orders of Magnitude

Scientific notation: a × 10^n	Used for numbers far from 1 in either direction — a unit cell (~0.3 nm = 3 × 10^{-10} m) or a fatigue life (10^7 cycles). a is between 1 and 10; n is the number of places the decimal point moves.
Multiplying/dividing in scientific notation	Multiply (or divide) the leading numbers normally, then add (or subtract) the exponents: $(2\times10^3)\times(3\times10^4) = 6\times10^7$.
Reading a logarithmic axis (the S-N curve)	Each gridline on a log-scale cycle axis represents ×10, not '+ some fixed amount' — the visual distance from 10^3 to 10^4 cycles is drawn the same as from 10^6 to 10^7, even though the second gap represents 900,000 more cycles than the first.
Why log scales are used for fatigue and grain size	Cycles to failure and grain diameters span many orders of magnitude (thousands to tens of millions of cycles; nanometers to millimeters) — a linear axis would squash the small values into an unreadable sliver, so a log scale is used to keep both ends legible.
Order-of-magnitude sanity checks	Before trusting a calculator answer, estimate the size by rounding each number to its leading digit and power of ten — if the calculator disagrees by more than a factor of ten from that estimate, a decimal point or exponent was likely entered wrong.

Rearranging Equations to Solve for an Unknown

The balance principle	An equation is a balance — whatever operation you perform on one side (add, subtract, multiply, divide, square, take a root) must be performed identically on the other side to keep it true.
Worked example: solve heat input for travel speed	$HI = (V\cdot A\cdot 60)/(1000\cdot s)$ → multiply both sides by $(1000\cdot s)$: $HI\cdot 1000\cdot s = V\cdot A\cdot 60$ → divide both sides by $(HI\cdot 1000)$: $s = (V\cdot A\cdot 60)/(HI\cdot 1000)$.
Worked example: solve turning roughness for feed	$Ra = f^2/(8r)$ → multiply both sides by 8r: $8r\cdot Ra = f^2$ → take the square root of both sides: $f = \sqrt{(8\cdot r\cdot Ra)}$.
Worked example: solve percent cold work for final area	$\%CW = (A_0 - A_f)/A_0 \times 100$ → divide by 100, multiply by A_0: $A_0\cdot(\%CW/100) = A_0 - A_f$ → rearrange: $A_f = A_0 - A_0\cdot(\%CW/100) = A_0(1 - \%CW/100)$.
Check the answer by substituting back in	After solving for an unknown, plug the result back into the original formula and confirm it reproduces the known value — the fastest way to catch an algebra mistake before it becomes a wrong shop measurement or a wrong exam answer.

Geometry & Right-Triangle Trigonometry

Basic Geometry Refresher

Angles around a point / a straight line	Angles on a straight line add to 180°; angles all the way around a point add to 360°. A right angle is 90°, marked with a small square in a diagram rather than a curved arc.
Complementary vs. supplementary angles	Complementary angles add to 90° (the two non-right angles in a right triangle are always complementary). Supplementary angles add to 180° (used when reading adjacent angles on a weld joint or a cut line).
Perimeter and circumference	Perimeter is the total distance around any shape. For a circle, circumference $C = \pi \cdot D = 2\pi \cdot r$, where D is diameter and r is radius — used to convert a bar's diameter into the length of stock consumed per revolution during turning.
Area of a circle	$A = \pi \cdot r^2 = \pi \cdot D^2/4$. This is the formula behind converting a measured rod or wire diameter into a cross-sectional area for a stress calculation ($\sigma = F/A$).
Area of a rectangle and a triangle	Rectangle: A = base × height. Triangle: A = ½ × base × height. Used for quick sanity-check estimates of a part's cross-section before a more exact calculation.
The Pythagorean theorem	For any right triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse (the side opposite the right angle) and a, b are the two legs. Rearranged forms: $c = \sqrt{(a^2 + b^2)}$, or $a = \sqrt{(c^2 - b^2)}$.
Worked example: Brinell hardness geometry	The Brinell formula's $D - \sqrt{(D^2 - d^2)}$ term is literally the Pythagorean theorem applied to the ball indenter: D is the ball diameter (the hypotenuse-like term), d is the diameter of the impression, and the difference gives the depth of penetration.

Right-Triangle Trigonometry — Sine, Cosine, Tangent

Labeling a right triangle	Pick one of the two non-right angles and call it θ. The hypotenuse is always the longest side, opposite the right angle. The 'opposite' side is across from θ; the 'adjacent' side is the other leg, next to θ.
SOH-CAH-TOA	sin(θ) = opposite/hypotenuse. cos(θ) = adjacent/hypotenuse. tan(θ) = opposite/adjacent = sin(θ)/cos(θ). These three ratios are fixed for a given angle no matter how large or small the triangle is.
Using sin, cos, tan to find a missing side	If θ and one side are known, multiply to find another side: opposite = hypotenuse × sin(θ); adjacent = hypotenuse × cos(θ); opposite = adjacent × tan(θ).
Using inverse trig to find a missing angle	If two sides are known, use the inverse functions to recover the angle: θ = $\sin^{-1}$(opposite/hypotenuse) = $\cos^{-1}$(adjacent/hypotenuse) = $\tan^{-1}$(opposite/adjacent). Most calculators label these asin, acos, atan or $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.
Common reference angles worth memorizing	30-60-90 and 45-45-90 triangles come up often as rules of thumb: sin(30°) = 0.5, sin(45°) = cos(45°) ≈ 0.707, sin(60°) ≈ 0.866. The 0.707 value in particular reappears throughout this course (see the fillet weld throat below).
Degrees vs. calculator mode	Every angle in this course is given in degrees, not radians — before using sin/cos/tan on a calculator, confirm it is set to 'DEG' mode, since the same keystrokes in 'RAD' mode return a completely different (wrong) number.

Course Applications — Where Trig Shows Up in the Shop

Fillet weld throat thickness	For a standard 45° fillet weld, throat = 0.707 × leg size (w). The 0.707 factor is exactly cos(45°) — the throat is the shortest distance from the weld root to the face, i.e., the side of a 45-45-90 triangle adjacent to the 45° angle formed by the two equal legs.
Angle-beam ultrasonic testing (UT) geometry	An angled UT probe sends sound along a slant path L into the part at a known refracted angle θ (commonly 45°, 60°, or 70°). The straight-line depth below the surface and the surface distance back to a reflector are both found with right-triangle trig once L and θ are known.
UT slant path from time of flight	$L = vS \cdot t/2$, where vS is the shear-wave velocity in the material and t is the round-trip travel time read off the flaw detector. Dividing by 2 accounts for the sound traveling to the reflector and back.
UT depth and surface distance from the slant path	Depth below the surface: $d = L \cdot \cos(\theta)$. Surface distance from the probe index point: $s = L \cdot \sin(\theta)$. θ is measured from the part surface's normal (perpendicular), so cos gives the 'straight down' component and sin gives the 'along the surface' component.
Worked example: 70° probe locating a flaw	Given vS = 3,230 m/s = 3.23 mm/μs and a measured travel time t = 18.5 μs: slant path L = 3.23 × 18.5/2 = 29.9 mm. Depth $d = L \cdot \cos(70°) = 29.9 \times 0.342 \approx 10.2$ mm below the surface. Surface distance $s = L \cdot \sin(70°) = 29.9 \times 0.940 \approx 28.1$ mm from the probe index point.
Draft and included angles in forming and machining	Draft angles on a die or mold, the included angle of a cutting tool tip, and the half-angle of a countersink are all just the θ in a right triangle relating a height (or depth) to a horizontal offset — the same sin/cos/tan relationships apply, just relabeled for the geometry at hand.

Engineering Equations & Key Concepts

Stress, Strain & Strengthening Relationships (Weeks 2, 7)

Engineering stress: $\sigma = F / A_0$	Internal force per unit original cross-sectional area. Units: Pa, MPa, ksi.
Engineering strain: $\varepsilon = \Delta L / L_0$	Change in length relative to original length — dimensionless, or expressed as %.
Hooke's Law: $\sigma = E \cdot \varepsilon$	Linear elastic region — E is the elastic (Young's) modulus, the slope of the stress-strain curve below the proportional limit.
0.2% offset method	Draw a line parallel to the elastic slope starting at $\varepsilon = 0.002$; its intersection with the curve is the reported yield strength when no natural yield point exists (e.g., aluminum).
$\%EL = (L_f - L_0)/L_0 \times 100\%$ · $\%RA = (A_0 - A_f)/A_0 \times 100\%$	Elongation at fracture and reduction in area — the two standard ductility measures from a tensile test.
True stress-strain (Hollomon equation): $\sigma = K \cdot \varepsilon^n$	Flow stress during plastic deformation beyond yield. K = strength coefficient, n = strain-hardening exponent — both material constants used throughout forming and machining analysis.
True strain: $\varepsilon_{true} = \ln(l / l_0)$	Unlike engineering strain, true strain accounts for the continuously changing gauge length/thickness — the correct measure once necking or heavy forming begins.
Hall-Petch relationship: $\sigma_\gamma = \sigma_0 + k / \sqrt{d}$	Yield strength increases as grain size (d) decreases — the practical basis for grain-refinement strengthening.
Taylor (dislocation) hardening: $\sigma \propto \sqrt{\rho}$	Strength scales with the square root of dislocation density — the mechanism behind work hardening.
Percent cold work: $\%CW = (A_0 - A_f)/A_0 \times 100\%$	Same form as reduction in area — quantifies how much a part has been cold-worked, which sets its strength/ductility per the %CW-vs-property curves and its recrystallization temperature on annealing.

Stress Concentration, Notches & Fatigue Design (cross-cutting — Weeks 2, 9, 11, 12, 15)

Stress concentration factor: $K_t = \sigma_{max} / \sigma_{nom}$	The theoretical (geometric) ratio by which a notch, hole, fillet, undercut, or sharp corner locally amplifies nominal applied stress — independent of material, a function of geometry alone (found from Peterson's charts or FEA).
Notch sensitivity: $q = (K_f - 1) / (K_t - 1)$	How much of the theoretical stress concentration a real material actually 'feels' in fatigue (0 = insensitive, 1 = fully sensitive). Brittle/high-strength materials tend toward $q \approx 1$; soft, ductile materials can locally yield and blunt the notch, giving $q < 1$.
Fatigue notch factor: $K_f = 1 + q(K_t - 1)$	The effective stress concentration factor for fatigue life prediction — always $\leq K_t$.
Real-world Kt magnitudes	Weld toes/roots: $K_t \approx 2$–5 (a major reason HIC and fatigue cracks concentrate there). Turbine cooling holes: $K_t \approx 2.8$. Even a fillet, keyway, or machining mark can produce a locally significant Kt — this is why 'the weld meets the drawing' does not guarantee fatigue performance.
Common manufacturing stress risers	Undercut, weld toe/root, sharp fillet radius, keyway corner, thread root, tool marks/chatter, porosity, inclusions, sudden section-thickness change — each one is a Kt source introduced (or avoided) by process control.
Residual stress: compressive vs. tensile	Compressive surface residual stress (shot peening, down/climb milling, cold rolling) delays fatigue crack initiation; tensile residual stress (grinding burn, welding, EDM recast layer) accelerates it — same applied load, very different fatigue life depending on which sign is present at the surface.
Mitigation methods	Generous fillet radii, smooth transitions, post-weld toe grinding or TIG dressing, shot peening, stress-relief PWHT — all exist specifically to lower Kt or convert tensile residual stress to compressive at the highest-Kt location.

Hardness Testing Formulas (Week 7 + Hardness Scales reference)

Rockwell: $HR = N - (h / s)$	N = scale constant, h = permanent depth increase (mm) after removing the major load, s = 0.002 mm/division. Higher number = shallower indent = harder material.
Brinell: $HB = 2F / [\pi D (D - \sqrt{D^2 - d^2})]$	F = load (kgf), D = ball diameter (mm), d = indentation diameter (mm). Standard test: 10 mm ball, F/D^2 held constant (30 for steel).
Vickers: $HV = 1.854 \times F / d^2$	F = load (kgf), d = mean diagonal of the square indentation (mm). One continuous scale from soft metals to diamond-like coatings.
UTS approximation from Brinell: UTS (MPa) $\approx 3.45 \times$ HB	Empirical, carbon/low-alloy steels only, valid roughly HB 100–400 — do not use for stainless or non-ferrous alloys.

Impact, Fatigue & Fracture Mechanics (Week 7 + Impact/Fatigue reference)

Charpy/Izod absorbed energy: $E = mg(h_0 - h_1)$	m = hammer mass, g = 9.81 m/s², h_0 = release height, h_1 = height reached after fracture. Reported in joules or ft-lb — higher = tougher.
Endurance limit approximation: $S_e \approx 0.5 \times$ UTS	For polished, unnotched steel specimens with UTS below ~1400 MPa. Above that, inclusions dominate and the ratio breaks down. Note this is the smooth-specimen value — apply Kf above to estimate the notched/as-manufactured endurance limit.
Fatigue ratio = S_e (or fatigue limit) ÷ UTS	Typically 0.4–0.5 for steels, 0.3–0.4 for aluminum alloys — useful for comparing materials at a glance.
S-N (Wöhler) curve	Stress amplitude vs. log(cycles to failure). Steels show a true horizontal endurance limit; aluminum and most non-ferrous alloys do not and must specify a life (e.g., 10^7 cycles).
Paris' Law: $da/dN = C(\Delta K)^m$	Governs Stage II fatigue crack growth rate per cycle as a function of the stress intensity factor range ΔK; C and m are material constants — this is the crack-growth counterpart to the Kt/Kf stress-concentration concepts above.

Manufacturing — Cutting Mechanics, Tool Life & Machining Parameters (Week 12 + Cutting/Turning-Milling references)

Cutting speed / spindle speed: $Vc = \pi \cdot D \cdot n / 1000$ · $n = 1000 \cdot Vc / (\pi \cdot D)$	D = workpiece or cutter diameter (mm), n = spindle speed (rpm). The most fundamental machining relationship — used to set up every turning and milling operation.
Material removal rate: $MRR = ap \cdot f \cdot Vc$ (turning) · $MRR = ae \cdot ap \cdot vf / 1000$ (milling)	ap = depth of cut, f = feed, vf = table feed rate, ae = radial engagement — the productivity metric that cutting parameter selection ultimately optimizes.
Merchant's shear angle: $\varphi = 45° + \alpha/2 - \lambda/2$	α = tool rake angle, λ = friction angle (= arctan μ). Higher rake or lower friction → larger shear angle → thinner chip → less cutting energy.
Chip thickness ratio: $r = t_1/t_2 = \sin\varphi / \cos(\varphi - \alpha)$	t_1 = uncut chip thickness, t_2 = actual chip thickness. Always r < 1 since the chip is thicker than the uncut layer (it must be, by volume conservation, once it shears and curls).
Specific cutting energy: $u = Fc / (t_1 \cdot b)$	Fc = cutting force, b = width of cut, in J/mm³. Depends on work material hardness and chip thickness (the 'size effect') — power = u × MRR.
Cutting force (Merchant's circle): $Fc = \tau s \cdot As / [\sin\varphi \cdot \cos(\lambda-\alpha)/\cos(\varphi+\lambda-\alpha)]$	τs = shear yield stress, As = shear-plane area = $t_1 \cdot b/\sin\varphi$. The thrust force component Ft acts perpendicular to Vc and can even reverse sign (pulling the tool into the work) at high positive rake.
Taylor's tool life equation: $Vc \cdot T^n = C$	T = tool life (min), n = Taylor exponent (0.1–0.4 HSS; 0.2–0.5 carbide), C = constant for the tool-material pair. Cutting speed is the single strongest driver of tool life — a modest Vc increase costs tool life exponentially.
Turning surface roughness: $Ra \approx f^2 / (8 \cdot r\varepsilon)$	f = feed per revolution, rε = tool nose radius. Finer feed and larger nose radius give a smoother surface — the same relationship introduced in Week 12, restated here alongside its milling counterpart.

Milling surface roughness: $Ra \approx fz^2/(8\cdot r\varepsilon)$ (flat end) · $Ra \approx ae^2/(8R)$ (ball end, scallop height)	fz = feed per tooth, R = ball radius, ae = radial step-over. Governs the theoretical finish achievable from tool geometry and feed alone, before chatter or tool wear are considered.
Chip thinning (milling): $hm = fz\cdot\sqrt{ae/Dc}$	When radial engagement ae is less than half the cutter diameter Dc, the actual chip is thinner than the programmed feed per tooth — feed must be increased ($fz\_adj = fz\_rec\cdot\sqrt{Dc/ae}$) to restore the intended chip load and avoid rubbing.
Cutting power: $Pc = Fc\cdot Vc / (60{,}000\cdot\eta)$	$\eta \approx$ machine efficiency (~0.80). Used to verify a cutting parameter set against available spindle motor power before committing to a program.
Heat partition in cutting	Roughly 75–80% of cutting heat leaves in the chip — why chip control and coolant matter for workpiece surface integrity (and why insufficient chip evacuation drives tool wear and workpiece thermal damage).

Manufacturing — Forming, Drawing & Sheet Metal (Weeks 10, 11 + Extrusion/FLD references)

Extrusion ratio: $ER = A_0 / A\_f$	Billet cross-sectional area divided by extrudate cross-sectional area. Typical range 10:1 to 100:1 for aluminum — a higher ratio demands more extrusion pressure and favors more extrudable (lower-strength) alloy series.
Recrystallization temperature: $Trex \approx 0.4 \cdot Tm$	Tm = absolute melting point. The dividing line between hot working (no net work hardening) and cold working (work hardening accumulates) for a given metal.
Strain-hardening exponent n and strength coefficient K	From $\sigma = K\cdot\varepsilon^n$. Higher n means a material spreads strain more uniformly before localizing into a neck — directly predicts formability in stretching operations.
Plastic strain ratio (Lankford coefficient): $\bar{r}$	Ratio of width strain to thickness strain in a tensile test — measures resistance to thinning. Higher $\bar{r}$ (e.g., ferritic stainless, some aluminum) means better deep-drawability.
Forming Limit Curve (FLC_0) estimate: $FLC_0 \approx (23.3 + 360\cdot t) \cdot n / 0.21$	t = sheet thickness (mm), n = strain-hardening exponent. Gives the plane-strain forming limit — the minimum point of the FLC and the critical value for most stamping operations.
Forming Limit Diagram (FLD) axes: major strain ε_1 (y-axis) vs. minor strain ε_2 (x-axis)	Both are true strains, $\varepsilon = \ln(l/l_0)$. Points below the FLC are safe; points above indicate through-thickness necking/fracture; points in the lower-left region ($\varepsilon_2 < -\varepsilon_1/2$) risk wrinkling instead.
Draw/reduction limits per pass	Wire/rod drawing: 15–25% area reduction per die. Tube drawing: 15–45% depending on method (sinking, fixed plug, floating plug, rod/mandrel) — exceeding these limits risks central bursts (chevron cracking) from excessive hydrostatic tension in the deformation zone.
Springback	Elastic recovery after a bend or draw releases — the part relaxes toward its original shape by an amount that depends on E, yield strength, and bend radius/thickness ratio; tooling must overbend to compensate.

Weldability & Heat Input (Weeks 4, 14, 15 + Preheat/ESAB references)

IIW Carbon Equivalent: CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15	The primary weldability index in AWS D1.1 — higher CE means higher hardenability and higher required preheat.
Pcm (Ito-Bessyo): Pcm = C + Si/30 + (Mn+Cu+Cr)/20 + Ni/60 + Mo/15 + V/10 + 5B	Better than CE for low-carbon, micro-alloyed/HSLA steels (<0.18% C) — gives a more accurate preheat estimate for those grades.
Heat Input: HI = (V × A × 60) / (1000 × travel speed) → kJ/mm	Higher heat input → slower cooling → coarser grains, wider HAZ, more distortion. Lower heat input → faster cooling → finer grains but higher HIC risk in hardenable steel.
Preheat threshold bands by CE	CE < 0.40: generally no preheat. CE 0.40–0.60: preheat ~100–200°C. CE > 0.60: preheat 200–300°C+, mandatory for most applications.
Dilution by process	SMAW 20–30% · GMAW 25–35% · SAW 50–70% — the fraction of deposited weld metal that is melted base metal; the WPS accounts for this when specifying filler composition.
Schaeffler diagram — Ni_eq = Ni + 30C + 0.5Mn	Nickel equivalent (austenite formers) used with a chromium equivalent to predict resulting phase balance (ferrite/austenite/martensite) in stainless steel weld metal.
WRC-1992 — Cr_eq = Cr + Mo + 0.7Nb, Ni_eq = Ni + 35C + 20N + 0.25Cu	A more accurate modern update to the Schaeffler diagram for duplex and modern stainless weld metal; predicts Ferrite Number (FN) — target 4–8 FN for hot-crack resistance without embrittlement.

Corrosion & Electrochemistry (Week 16)

PREN = %Cr + 3.3 × %Mo + 16 × %N	Pitting Resistance Equivalent Number — ranks stainless/nickel alloys by chloride pitting resistance. 304 ≈ 19, 316 ≈ 24, 2205 duplex ≈ 35–40.
PREN thresholds	<18 low resistance · 18–25 moderate (fresh water) · 25–35 good (brackish/marine) · 35–40 high (seawater) · >40 very high (full seawater/desalination).
Cathodic protection criterion: −0.85 V vs. $Cu/CuSO_4$	Per NACE SP0169 — the minimum structure-to-electrolyte potential considered adequately protected.
Corrosion rate: mm/year or mils per year (MPY)	The standard way uniform corrosion rate is measured and used to calculate a corrosion allowance and remaining service life.

NDT Physics (Week 8 + NDT Methods reference)

Eddy current skin depth: $\delta = 503 / \sqrt{f \times \mu_r \times \sigma}$	f = frequency (Hz), μ_r = relative magnetic permeability, σ = conductivity (MS/m). Higher frequency = shallower, more surface-sensitive inspection; lower frequency reaches deeper but loses surface sensitivity.